SHALIAH FRICAS

Boston, Massachusetts | 470-865-8545 | shy.fricas30@gmail.com
Portfolio | linkedin.com/in/shaliahfricas | GitHub | ORCID: 0009-0007-8990-8371

Research Interests

Human-centered robotic systems; biomedical and assistive robotics; trustworthy autonomous systems; sensor reliability and resilient navigation; cybersecurity for cyber-physical systems.

Education

University of Massachusetts Boston Expected May 2027
B.S. in Computer Engineering | Minor in Computer Science
Selected coursework: Embedded Systems; Artificial Intelligence; Digital Systems; Data Structures & Algorithms; Circuit Analysis.
Planned Spring 2027 coursework: CS 666—Biomedical Signal & Image Processing; ENGIN 478—Independent Study (3 credits).

Research Experience

TRIO McNair Scholars Program — ICARUS Lab, UMass Boston February 2026–Present
Undergraduate Researcher, Robotics & Cyber-Physical Systems / Mentor: Prof. Ioannis Zografopoulos

- Configured a TurtleBot3 Burger ROS 2 Humble research platform with LiDAR, Cartographer mapping, AMCL localization, and Nav2 autonomous navigation to establish a repeatable physical baseline.
- Investigating autonomous navigation under correct versus incorrect or manipulated sensor input, with emphasis on measurable reliability, fault response, and security implications.
- Proposed **ENGIN 478 Independent Study**, *Sensor Reliability and Resilient Navigation in Autonomous Robotic Systems*, as a Spring 2027 continuation of the McNair research; planned components include IoT system architecture, telemetry, machine learning for sensor data, fault injection, and OPAL-RT simulation.

Research GitHub

UCaN Lab, UMass Boston September 2025–Present
Undergraduate Researcher, Wireless Communications & Machine Learning / Advisor: Prof. Rahaim

- Developed MATLAB BiLSTM classification experiments using software-defined-radio in-phase/quadrature (I/Q) data.
- Investigated hybrid radio-frequency (RF), optical wireless communication (OWC), and visible-light positioning (VLP) approaches for localization in GPS-denied environments.

Engineering & Professional Experience

Operation Autopilot, Inc. — Robotics Field Engineer January 2026–Present
Boston, Massachusetts

- Diagnose and troubleshoot autonomous mobile robots deployed in customer environments, including localization, navigation, docking, charging, and task-execution faults.
- Configure navigation maps, boundaries, return points, schedules, and robotic tasks; validate corrective actions through repeat testing and document technical findings.

University of Massachusetts Boston — Service Desk Analyst Present
Information Technology

- Provide technical assistance and troubleshoot computing and IT issues in a university service-desk environment.

Best Buy — Certified Apple Computing Advisor June 2023–January 2025

- Troubleshot computing devices and supported customers with technical issues, Apple-product guidance, and practical computing advice.

NeuralSeek — AI Agent Builder Intern August–September 2025
Remote (Miami, Florida)

- Collaborated on generative-AI agents for healthcare, human resources, and business workflows; translated use cases into agent behavior and checked responses against requirements.

Arka HR — Operations & App Development Intern June 2023–December 2024
Boston, Massachusetts

- Streamlined internal workflows through data organization, process improvements, and support for cross-team application-development activities.

Selected Engineering Projects

AYDR Node — Senior Design Secure Robotics Gateway 2026–Present

PII isolation, security verification, and embedded integration *Project Files*

- Developing a plug-and-play security gateway for robotic and IoT systems, incorporating authenticated commands, revocation checks, safety constraints, behavioral anomaly monitoring, secure-element identity, and tamper-evident logging.
- Contributing to private-information isolation, verification design, and ESP32/Raspberry Pi integration for a two-arm robotic demonstration.

AFRL / HyLo SDR Indoor Localization Project — Finalist Team 2025–2026

Machine-learning signal classification *Project Files*

- Built MATLAB BiLSTM signal-classification experiments with SDR I/Q data for a hybrid RF/OWC/VLP localization approach in GPS-denied environments.

Modular Autonomous Underwater Vehicle (AUV) — Engineering & Robotics Club 2026–Present

Autonomous systems and simulation *Project GitHub*

- Contributed to robotics-club development around underwater-vehicle simulation, mission concepts, interface planning, and system integration.

MediSeek — Biomedical AI Troubleshooting Agent 2025

- Developed an AI-assisted troubleshooting workflow designed to help biomedical engineers and healthcare teams organize medical-device fault diagnosis. Project Files

Favorite Engineer — Free Engineering Interview Study Tool 2026

- Built and published a free web-based study resource to help engineering students prepare for technical interviews and review engineering topics. Live Website

Research Presentations & Competitions

MassURC Undergraduate Research Symposium — Research Poster Presenter 2026

Presented undergraduate research to faculty and student audiences.

AFRL / HyLo SDR Indoor Localization Competition — Finalist 2026

Finalist-team recognition for hybrid localization research integrating software-defined radio and optical/radio techniques.

Leadership & Service

UMass Boston CyberSecurity Club — Event Planner, Executive Officer Fall 2026–Present

- Accepted an officer role following connections made through McNair research; support event planning and club operations alongside the president and treasurer.

Vice Chancellor for Student Affairs' Undergraduate Leadership Council 2026–2027

Student Representative, University of Massachusetts Boston

- Serve as a student representative contributing perspectives on campus experience and student-affairs priorities.

Engineering & Robotics Club — MassRobotics Competition Team 2026–Present

- Participate in collaborative robotic-system development and competition activities.

Order of the Engineer Initiative — Student Organizer In Progress

- Initiated conversations with university engineering leadership about establishing an Order of the Engineer presence on campus.

Honors & Professional Development

- TRIO McNair Scholars Program — Undergraduate Research Fellow.
- AFRL / HyLo SDR Indoor Localization Competition — Finalist team.
- Oxford Artificial Intelligence Programme — Certificate of Completion, Saïd Business School, University of Oxford (October 2025).

Technical Skills

Programming: Python, C, C++, MATLAB, Bash.
Robotics & Embedded: ROS 2, Nav2, TurtleBot3, microcontroller programming and interfacing (ESP32), Raspberry Pi, LiDAR, autonomous navigation, sensor integration, OPAL-RT real-time simulation / HIL.
Signals & Engineering Tools: Linux/Ubuntu, Git/GitHub, GNU Radio, RTL-SDR, RF, OWC/VLP, KiCad, LTspice, CAD, VS Code, Overleaf/LaTex.
Languages: English and Spanish (bilingual).